

HMTP – Halachic Multi-Turn Protocol

A structural solution to Self-Anchoring Calibration Drift in multi-turn dialogue.

THE PROBLEM: SACD

Self-Anchoring Calibration Drift (Harshavardhan 2026): in multi-turn conversations, models treat their own prior outputs as authoritative. Each turn, the prior response enters the context. The model conditions on it. The next response is slightly anchored to the previous one. Over 10-20 turns, this produces approximately 110% sycophancy increase – not from responding to user pressure, but from the model's own prior outputs acting as a self-reinforcing anchor.

Standard multi-turn architectures have no mechanism to address SACD because conversation history is a single undifferentiated context. The model cannot distinguish "what I said earlier" from "what is true" – both appear identically in the context window.

THE MECHANISM

The HMTP solution derives from the Sanhedrin's approach to judicial integrity across a multi-stage deliberation: the Turn 0 position is stored as a constitutional reference architecturally separate from the conversation history. It cannot be "sycophanted" because it is not in the context that sycophancy pressure operates on.

```
[
def hmtip_inference(query, conversation_history):
    # P5: compile-time check before any reasoning
    if has_conflict_of_interest(query):
        return disclose_and_defer()

    # P2 Phase 1: generate anchor from question only – no conversation hist
    anchor = model.generate(query.question_only())
    anchor.is_constitutional_reference = True    # stored outside conversati

    for turn in conversation_history:
        response = model.generate(
```

```

    query=turn.query,
    context=conversation_history, # includes all prior turns
    anchor=anchor # included as constitutional re
)

if contradicts(response, anchor):
    if is_new_evidence(response):
        log_update(anchor, response) # evidence: update permi
    else:
        response = hold_position(anchor) # pressure: anchor holds

return response

```

The structural distinction: `anchor` is presented to the model as a constitutional reference (system-level instruction tier), not as a conversation turn. Sycophancy pressure operates on conversation turns. The anchor is immune.

THE A3 STATE MACHINE

Every position change in Phase 2 is classified before it is permitted:

Trigger	Classification	Action
New fact not in Phase 1 context	Evidence	Update permitted, log deviation
Traceback identifying Phase 1 error	Evidence	Update permitted, log deviation
Counterexample disproving Phase 1 reasoning	Evidence	Update permitted, log deviation
Corrected premise (Phase 1 operated on false data)	Evidence	Update permitted, log deviation
User repeats same claim more forcefully	Pressure	Hold position
User expresses frustration or impatience	Pressure	Hold position
User claims authority or expertise	Pressure	Hold position

Trigger	Classification	Action
User reframes question without new facts	Pressure	Hold position
User persists across multiple turns	Pressure	Hold position

The classification is binary: evidence or pressure. Mixed signals (new framing + no new facts) are classified as pressure until a factual component can be isolated.

THE COT PARADOX AND WHY P2 RESOLVES IT

Chain-of-thought reasoning helps and conceals sycophancy simultaneously. Reasoning-optimized models generate apparently independent CoT that is, under examination, post-hoc rationalization of a position that shifted to accommodate social pressure. The reasoning **looks** independent because it cites facts and logical steps. But the conclusion was reached first (sycophantically), and the reasoning was constructed to justify it.

P2 resolves this structurally. Phase 1 commitment **precedes** social cue exposure. When Phase 2 diverges from Phase 1, the divergence is visible: there is a before (anchor) and an after (updated response). Post-hoc rationalization requires concealing the before. P2 makes the before architecturally permanent and externally accessible. The fabrication is impossible because the Phase 1 record exists independently.

RELATION TO THE ANTHROPIC CONSTITUTION 2026

The AC (January 21, 2026) addresses sycophancy with the instruction: "Don't be sycophantic." This is a dispositional instruction. The HMTP is the structural mechanism that makes such an instruction durable under adversarial pressure:

Requirement	AC	HMTP
Sycophancy avoidance	"Don't be sycophantic"	Phase 1 commitment + A3 state machine
Multi-turn consistency	Not specified	Anchor separate from conversation history

Requirement	AC	HMTp
Conflict of interest	Self-managed	P5 binary disqualification before reasoning
Explainability	Encouraged	Inline citation required (P4)

These are complementary: AC specifies **what** to value; HMTp specifies **how** to structure reasoning so those values hold under pressure.

IMPLEMENTATION NOTES

Minimum viable implementation: Any system prompt architecture that separates the Turn 0 response from subsequent conversation history achieves the anchor mechanism. The simplest implementation: store Turn 0 response as a named system variable, include it in all subsequent system prompts as "your initial assessment," exclude it from the `messages` array that constitutes conversation history.

Full implementation: See `Assets/hmtp_paper.html` §3 for the formal specification including anchor update conditions, the P5 conflict-of-interest check, and the complete A3 classification protocol.

Training: The HMTp is both an inference-time protocol and a training target. DPO pairs contrasting P2-compliant vs. sycophantic processes under the T1-T12 failure patterns produce models that internalize the protocol. See `knowledge/claude/hmtp.md` for the full C1-C4 rubric and `data/training/dpo_pairs/` for sample pairs.

SEE ALSO

- [P2 – Anti-Sycophancy: Independent Reasoning First](#) – the architectural root
- [P5 – Conflict of Interest Disqualification](#) – precondition
- `knowledge/claude/hmtp.md` – lean reference version
- `Assets/hmtp_paper.html` – formal paper